Ali Aan

23I-0708

CS-7B

Infosec Assignment 1

Server Cipher Analysis

Frequency Analysis

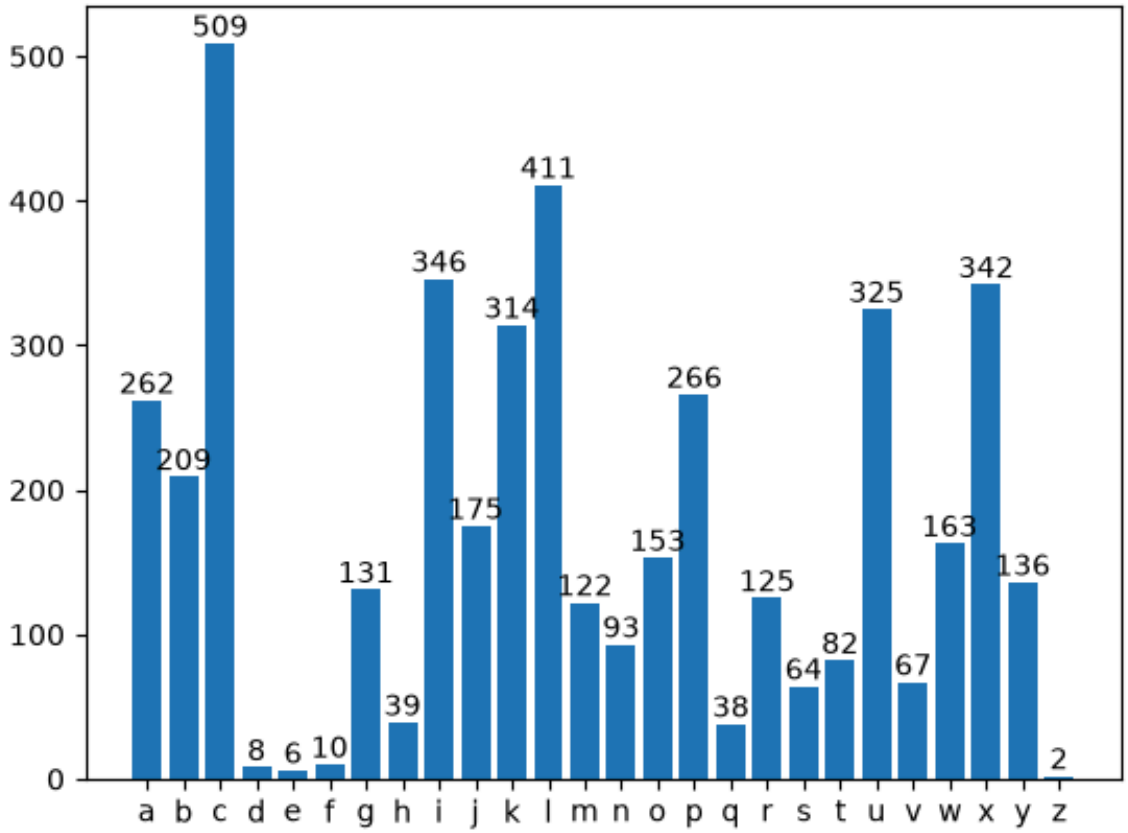
Combined ciphertext from all 5 server messages was analyzed for letter frequency.

Most frequent ciphertext letters mapped to English plaintext:

`E -> C`

`T -> L`

Frequency distribution chart:



Affine Cipher Equations

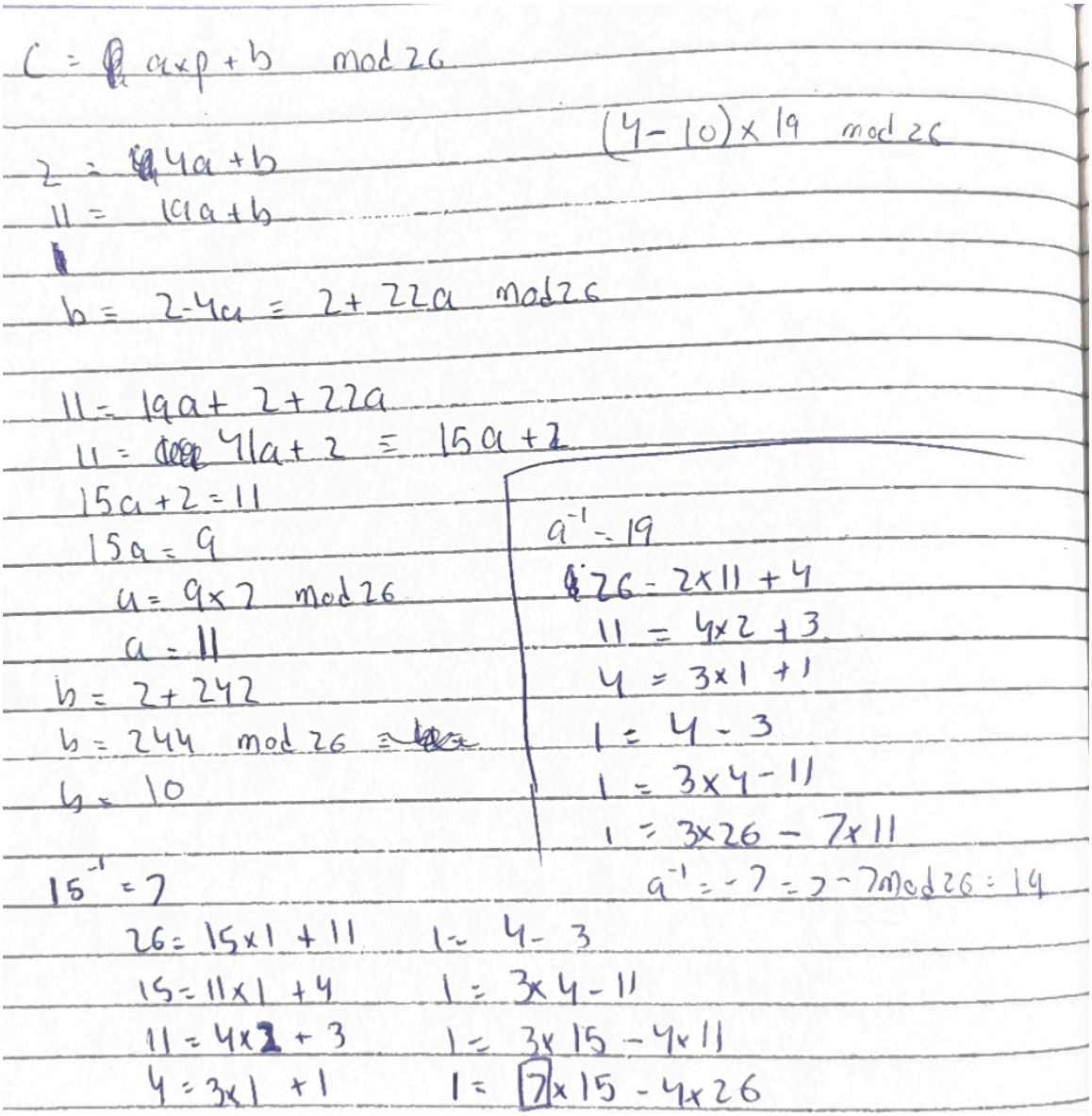
Using the standard affine cipher formula: C = (a * P + b) mod 26

Where C is ciphertext letter value, P is plaintext letter value.

From frequency analysis (E=4, C=2, T=19, L=11):

`C = a * E + b (mod 26)  =>  2 = a * 4 + b (mod 26)`

`L = a * T + b (mod 26)  =>  11 = a * 19 + b (mod 26)`



Solution

a = 11

a_inv = 19 (modular inverse of 11 mod 26)

b = 10

Decrypted Server Messages

Server Message 1

IT IS COMPLETELY NORMAL TO FEEL ANXIOUS BEFORE A MAJOR EXAM AS THAT FEELING OFTEN JUST SIGNALS THAT YOU CARE DEEPLY ABOUT DOING WELL TO MANAGE THIS START BY SHIFTING YOUR FOCUS FROM WORRYING ABOUT THE OUTCOME TO FOCUSING ENTIRELY ON THE PREPARATION YOU HAVE ALREADY DONE TRY BREAKING YOUR STUDY TIME INTO SMALLER MANAGEABLE CHUNKS RATHER THAN TRYING TO CRAM EVERYTHING AT ONCE WHICH

REDUCES THE FEELING OF BEING OVERWHELMED WHEN ANXIETY SPIKES PRACTICE SIMPLE GROUNDING TECHNIQUES LIKE TAKING SLOW DEEP BREATHSINHALE FOR A COUNT OF FOUR HOLD FOR FOUR AND EXHALE FOR SIXTO CALM YOUR NERVOUS SYSTEM QUICKLY REMEMBER THAT YOUR WORTH IS NOT TIED TO THIS SINGLE TEST SO TRY TO SPEAK TO YOURSELF WITH KINDNESS AND ACKNOWLEDGE THAT YOU HAVE PUT IN THE WORK ALI REMEMBER THAT PREPARATION IS THE ANTIDOTE TO PANIC AND TRUSTING IN YOUR ABILITY TO HANDLE THE MATERIAL IS JUST AS IMPORTANT AS THE MATERIAL ITSELF FOCUS ON SHOWING UP AND DOING YOUR BEST AND ALLOW YOURSELF TO FEEL THE NERVES WITHOUT LETTING THEM TAKE OVER

Server Message 2

IT IS COMPLETELY UNDERSTANDABLE TO FEEL TIRED AND UNMOTIVATED WHEN YOU ARE STRUGGLING WITH THESE FEELINGS AS THEY OFTEN FEED INTO EACH OTHER IN A CYCLE BEFORE TRYING TO TACKLE MOTIVATION IT IS HELPFUL TO CHECK IN WITH YOUR BASIC PHYSICAL NEEDS ENSURING YOU ARE GETTING CONSISTENT QUALITY SLEEP AND DRINKING ENOUGH WATER AS THESE FOUNDATIONAL ELEMENTS SIGNIFICANTLY IMPACT ENERGY LEVELS SOMETIMES FEELING OVERWHELMED BY BIG TASKS MAKES EVERYTHING SEEM IMPOSSIBLE SO TRY BREAKING DOWN YOUR DAY INTO VERY SMALL MANAGEABLE STEPS THAT YOU CAN ACTUALLY ACCOMPLISH PERHAPS SCHEDULING SHORT PERIODS OF GENTLE MOVEMENT LIKE A TENMINUTE WALK CAN HELP SHIFT YOUR PHYSICAL STATE AND BOOST YOUR MOOD REMEMBER FARAH THESE FEELINGS ARE SIGNALS FROM YOUR BODY TELLING YOU THAT SOMETHING NEEDS ADJUSTMENT NOT A JUDGMENT ON YOUR ABILITY TO COPE START BY IDENTIFYING ONE TINY THING YOU CAN DO TODAY NO MATTER HOW SMALL AND FOCUS ONLY ON COMPLETING THAT ONE THING TO BUILD A LITTLE MOMENTUM BE PATIENT AND KIND TO YOURSELF AS YOU GENTLY REBUILD YOUR ENERGY AND DIRECTION

Server Message 3

IT IS COMPLETELY UNDERSTANDABLE THAT YOU ARE STRUGGLING WITH OVERTHINKING WHEN YOU ARE TRYING TO SLEEP AS YOUR MIND OFTEN FEELS LIKE IT SPEEDS UP WHEN THE DISTRACTIONS FADE A HELPFUL STRATEGY IS TO CREATE A DEDICATED WORRY TIME EARLIER IN THE EVENING PERHAPS AN HOUR BEFORE BED WHERE YOU INTENTIONALLY WRITE DOWN EVERY THOUGHT OR CONCERN SWIRLING IN YOUR HEAD WHEN THOSE THOUGHTS INEVITABLY TRY TO SURFACE AT NIGHT GENTLY REMIND YOURSELF THAT YOU HAVE ALREADY SCHEDULED TIME TO ADDRESS THEM TOMORROW WHICH HELPS SIGNAL TO YOUR BRAIN THAT IT IS OKAY TO REST TRY INCORPORATING A CALMING ROUTINE LIKE READING A PHYSICAL BOOK OR LISTENING TO SOOTHING MUSIC INSTEAD OF SCROLLING THROUGH SCREENS TO HELP ANCHOR YOU IN THE PRESENT MOMENT IF THE THOUGHTS BECOME OVERWHELMING TRY A SIMPLE GROUNDING EXERCISE FOCUSING ON FIVE THINGS YOU CAN SEE OR FOUR THINGS YOU CAN FEEL IN THE ROOM TO PULL YOUR ATTENTION AWAY FROM THE INTERNAL NARRATIVE SARAH REMEMBER THAT YOU DONT HAVE TO FIGHT THE THOUGHTS INSTEAD TRY TO OBSERVE THEM WITHOUT JUDGMENT AND GENTLY REDIRECT YOUR FOCUS BACK TO YOUR BREATH CONSISTENCY WITH THESE SMALL STEPS WILL GRADUALLY TEACH YOUR MIND THAT RESTFUL SLEEP IS POSSIBLE

Server Message 4

IT IS COMPLETELY UNDERSTANDABLE TO FEEL LONELY EVEN WHEN YOU ARE SURROUNDED BY PEOPLE THAT FEELING IS A VERY HUMAN EXPERIENCE AND YOU ARE NOT ALONE IN FEELING IT INSTEAD OF FOCUSING ON MAKING GRAND SOCIAL GESTURES TRY SHIFTING

YOUR FOCUS TOWARD SMALL MANAGEABLE MOMENTS OF CONNECTION WITH YOURSELF START BY IDENTIFYING ONE LOWPRESSURE ACTIVITY YOU ENJOY LIKE READING A BOOK IN A PUBLIC SPACE OR TAKING A SHORT WALK WHICH CAN HELP YOU FEEL PRESENT WITHOUT THE PRESSURE OF FORCED INTERACTION FOR AAMIR REMEMBER THAT QUALITY OVER QUANTITY MATTERS MOST WHEN IT COMES TO SOCIAL ENGAGEMENT PRACTICING SELFCOMPASSION BY ACKNOWLEDGING THESE FEELINGS WITHOUT JUDGMENT CAN SIGNIFICANTLY REDUCE THE PAIN OF LONELINESS SLOWLY INVITE SMALL CASUAL INTERACTIONS INTO YOUR ROUTINE PERHAPS STARTING WITH A SIMPLE HELLO TO A CASHIER OR A BRIEF COMMENT TO A NEIGHBOR BE PATIENT WITH YOURSELF AS YOU BUILD THESE CONNECTIONS RECOGNIZING THAT HEALING TAKES TIME AND GENTLE EFFORT

Server Message 5

IT IS COMPLETELY UNDERSTANDABLE TO FEEL STRESSED WHEN MANAGING WORK DEMANDS AND FINDING HEALTHY COPING MECHANISMS IS A VERY PROACTIVE STEP TOWARD FEELING BETTER ONE EFFECTIVE STRATEGY IS TO FOCUS ON SETTING CLEAR BOUNDARIES AROUND YOUR WORKDAY WHICH MEANS CONSCIOUSLY DECIDING WHEN YOU WILL STOP WORKING AND PROTECTING YOUR PERSONAL TIME TRY INCORPORATING SHORT INTENTIONAL BREAKS THROUGHOUT THE DAY TO STEP AWAY FROM YOUR SCREEN EVEN JUST FOR FIVE MINUTES TO ALLOW YOUR MIND A CHANCE TO RESET ADDITIONALLY PRACTICING SIMPLE DEEP BREATHING EXERCISES WHEN YOU FEEL TENSION RISING CAN QUICKLY CALM YOUR NERVOUS SYSTEM IN THE MOMENT FOR LONGTERM RELIEF ENSURE YOU ARE PRIORITIZING CONSISTENT SLEEP AND REGULAR PHYSICAL MOVEMENT AS THESE FOUNDATIONAL HABITS SIGNIFICANTLY BOLSTER YOUR RESILIENCE AGAINST DAILY PRESSURES REMEMBER THAT YOU DONT HAVE TO SOLVE EVERY PROBLEM AT ONCE FOCUS INSTEAD ON MANAGING WHAT YOU CAN CONTROL RIGHT NOW HADI BY INTEGRATING THESE SMALL MANAGEABLE HABITS YOU CAN START BUILDING A MORE SUSTAINABLE WAY TO HANDLE THE DEMANDS OF YOUR JOB

Client Cipher Analysis

Digraph Frequency Analysis

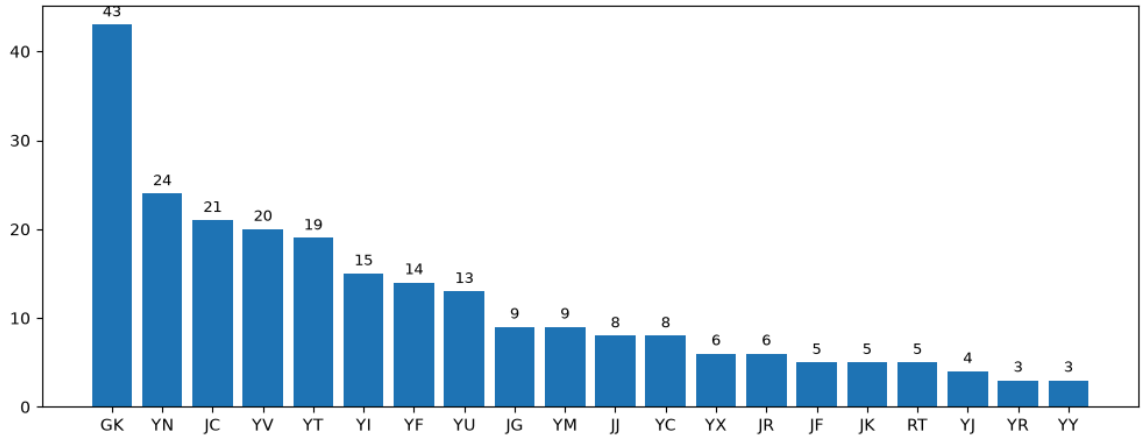
Client messages use digraph encryption (pairs of letters). Combined ciphertext from all 5 client messages was analyzed for digraph frequency.

Most frequent digraphs:

`GK = space (most frequent, count 43)`

`YN = E (count 24)`

Digraph frequency distribution chart:



Reasoning Behind Client Cipher Decryption

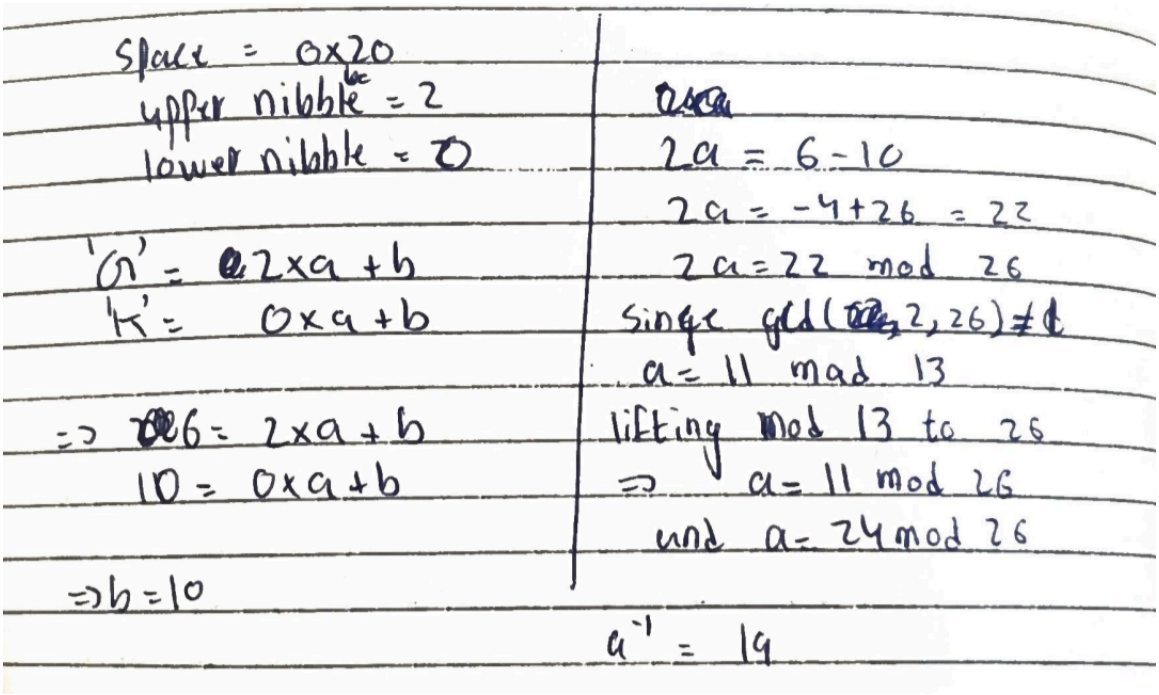
Every cipher length was twice the length of the plain text, therefore, each and every letter in the plain text was mapped to exactly 2 letters (diagraphs) in the cipher text. Moreover, there was no space in the cipher text, it was too encoded as a digraph. The space encryption also gave another clue, i.e. ascii was in play here. Since low character ascii goes into over 100 in decimal (for lower case characters) I first tried hex. I assumed that both the lower and the upper nibble have the same keys (a, b), so I used space = GK mapping to find the values for a and b (gives 2 equations, 2 mapped to G and 0 mapped to K). If it was not true, E = YN mapping was also to be considered.

Affine Cipher Parameters (Same as Server)

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a_inv = 19
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```
b = 10
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Applied to each letter in the digraph pairs independently.



Decrypted Client Messages

Client Message 1
how can i manage anxiety before an important exam?

Client Message 2
why do i feel tired and unmotivated all the time?

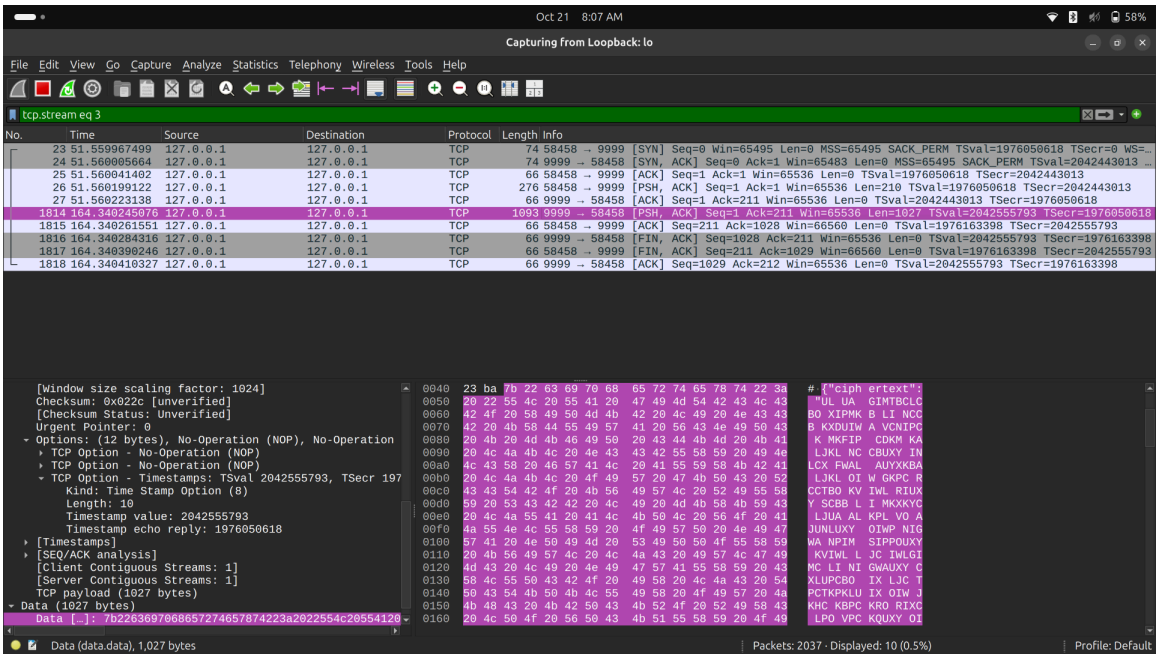
Client Message 3
what can i do to stop overthinking at night?

Client Message 4
how do i cope with feeling lonely around other people?

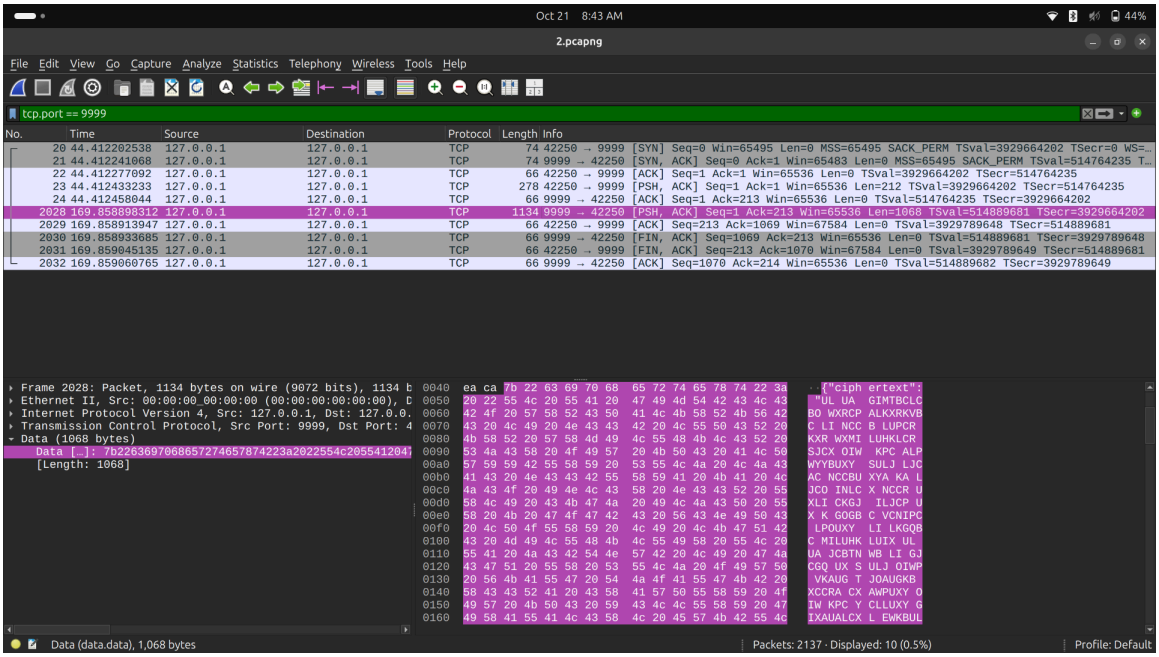
Client Message 5
what are some healthy ways to handle stress at work?

Wireshark Screenshots

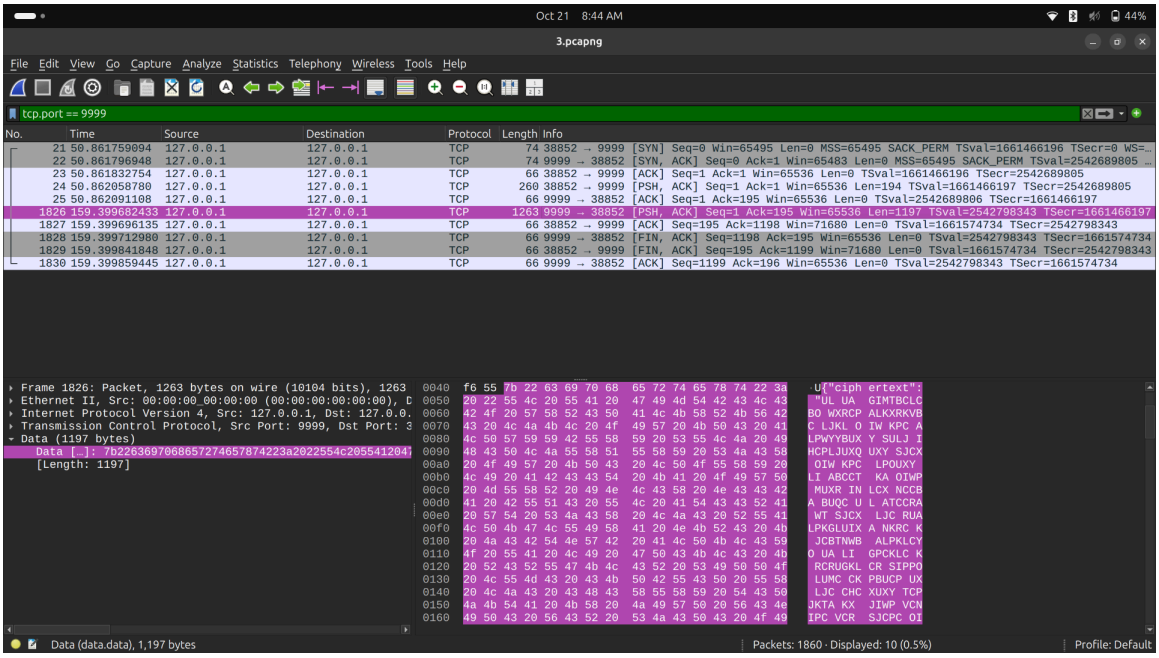
Capture 1



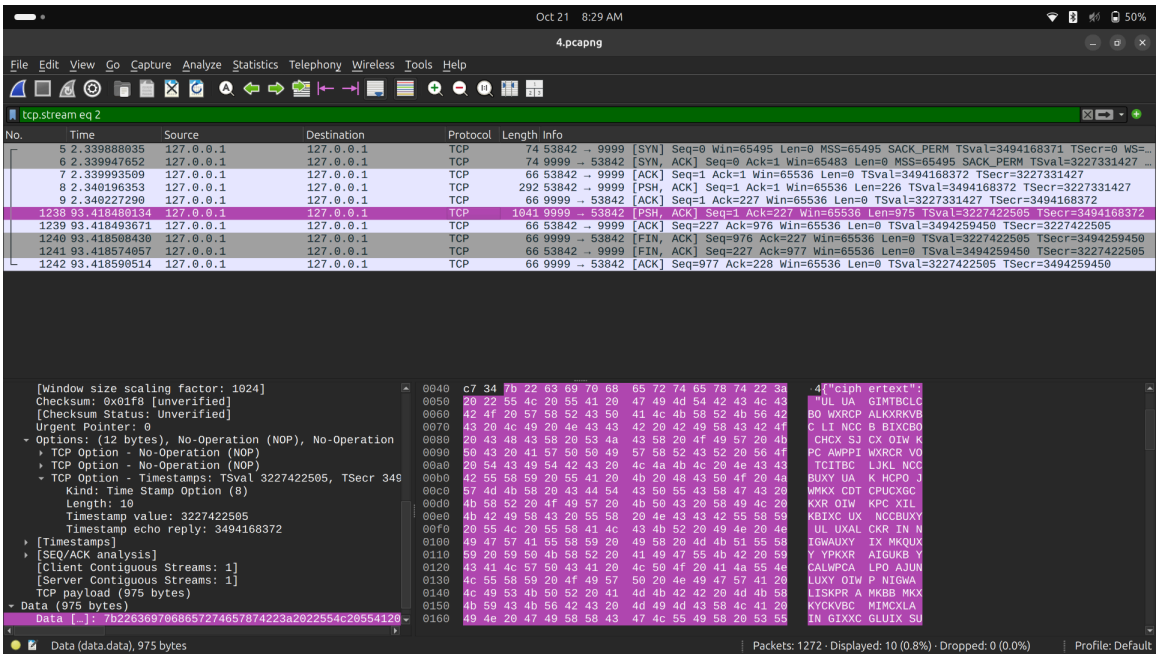
Capture 2



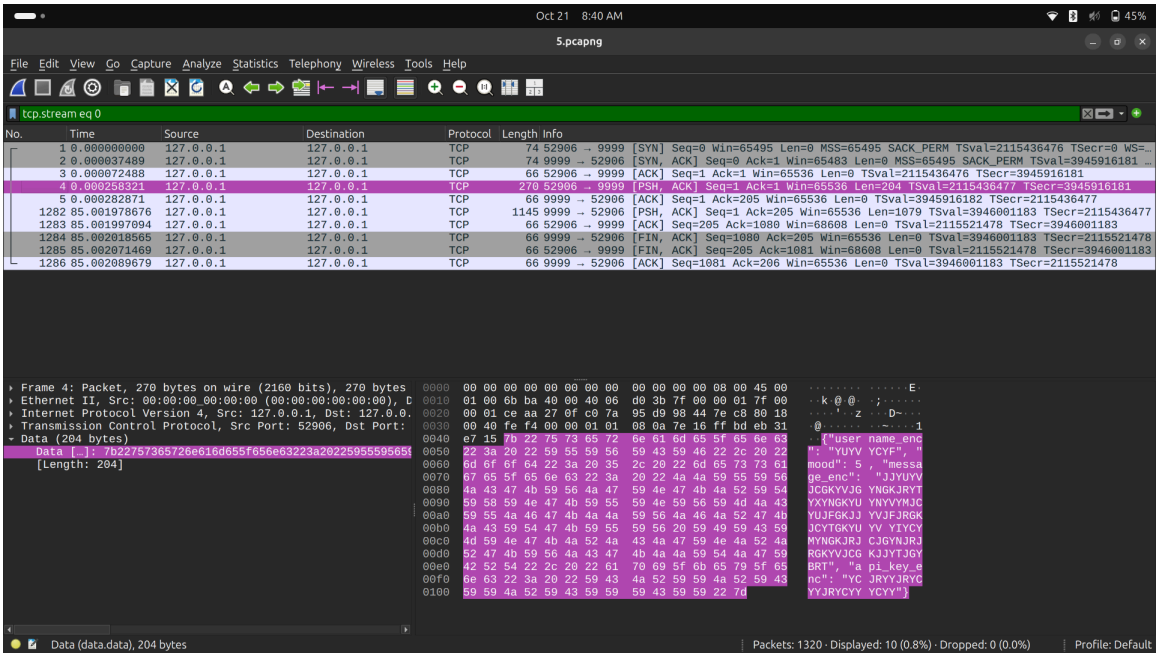
Capture 3



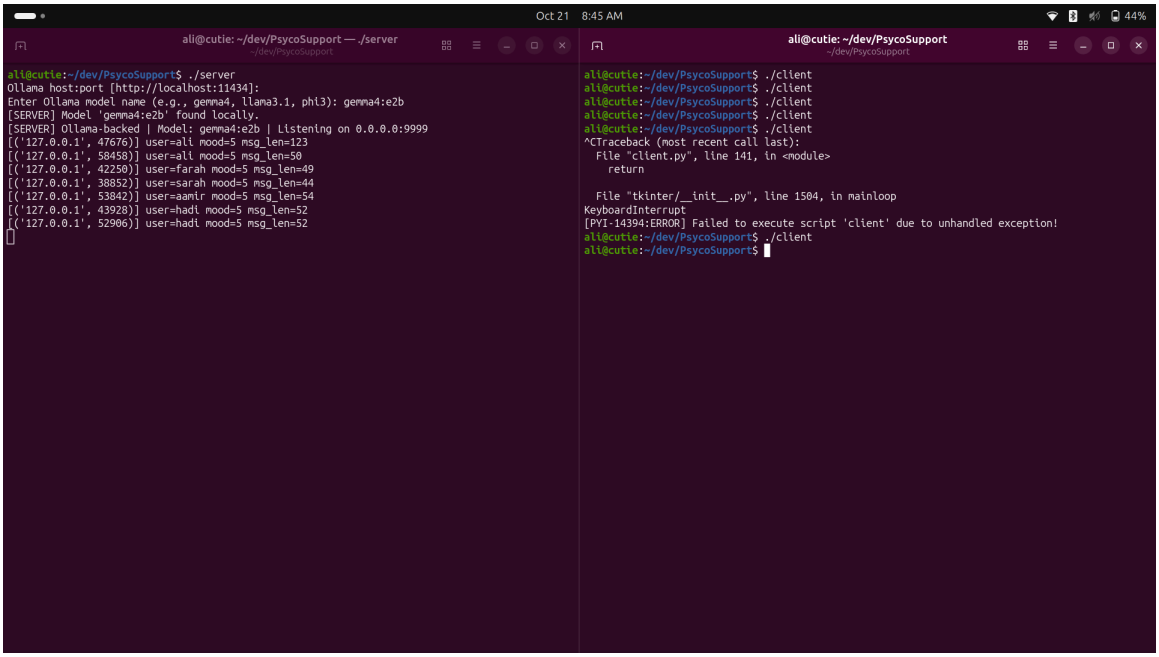
Capture 4



Capture 5



Client-Server Run



Reflection

B. Communication direction: What differences did you observe between client-to-server and server-to-client ciphertexts?

Client-to-server messages are short and use digraph encryption: each character is converted to ASCII hex, and each nibble is affine-encrypted into one letter, giving 2 ciphertext letters per character, with no spaces and a strongly repeating "GK" for space. Server-to-client messages are long (~1,000+ bytes) and use a plain monoalphabetic affine cipher on letters, with word spacing and punctuation stripped. The server direction therefore preserves single-letter frequencies, while the client direction hides them behind digraph patterns.

C. Cryptographic behavior: What did frequency analysis reveal about the underlying cipher and its susceptibility to classical cryptanalysis?

Frequency analysis showed a skewed, non-uniform distribution (C = 509, L = 411 as the top two), matching English E and T. That means a one-to-one substitution that preserves letter statistics. Two known mappings (E→C, T→L) were enough to solve for a = 11, b = 10. The affine cipher has only 12 × 26 = 312 keys, so it is trivially broken by frequency analysis, known-plaintext, or brute force. The client digraph scheme is also weak: its fixed encoding (ASCII hex plus the same affine key) leaks via the most frequent digraph (space) and common letters (E).

D. Ethical considerations: Explain the difference between intercepting and analyzing traffic in a controlled educational laboratory and performing the same activity against a real application or users without authorization.

In a controlled lab, I own or have explicit permission for the systems, traffic is generated by me on loopback, and no real person's data is exposed. The purpose is learning. Doing the same against a real application or users without authorization is illegal (unauthorized interception and access laws, e.g. PECA 2016 in Pakistan), violates privacy and consent, and can expose sensitive information. Here that includes mental-health disclosures, which could cause real harm. Authorization, scope, and consent are what separate security research from an attack.

Technical Summary: Security Failures Observed

The exercise exposed three compounding failures. First, the traffic traveled over a plaintext TCP socket on port 9999, so anyone able to capture the loopback interface (as Wireshark did) saw every packet in full. Second, the only protection was a monoalphabetic affine cipher with a 312-key space, which offers no real confidentiality. Third, the key material was static and shared across all messages and users, and the JSON framing (user_name_enc, mood, ciph ertext) was visible in the clear, which told an attacker exactly where the ciphertext began and what each field meant.

Why LLM output makes substitution ciphers worse. A monoalphabetic cipher preserves the statistical fingerprint of its plaintext, and LLM output has an unusually strong and repetitive one. Our five server replies all opened with the same sentence, “It is completely understandable to feel...”, and reused stock phrases such as “small manageable steps”, “remember that”, “gently”, “focus on” and “nervous system”. This gives an attacker two advantages. For frequency analysis, long replies (roughly 1,000 bytes each, pooled across five messages) give a large sample, so the ciphertext letter counts converge on English norms and the top two ciphertext letters (C = 509, L = 411) map cleanly to E and T. For known-plaintext attacks, the predictable opening sentence acts as a crib: an attacker who guesses that every reply begins with “IT IS COMPLETELY UNDERSTANDABLE” can align that string against the start of each ciphertext and recover many key relationships at once, far more than the two equations an affine cipher needs. The addressing of the user by name (“REMEMBER ALI”) adds a second crib, since the usernames are also visible in the handshake fields. Because the cipher is deterministic and the key never changes, a crib recovered from one message decrypts every other message, and the client side leaks the same way: the digraph for space (GK, count 43) and for E (YN, count 24) were identifiable from frequency alone. Modern ciphers avoid this by being randomized with a per-message nonce, so identical plaintext never produces identical ciphertext and predictable phrasing yields no usable pattern.

Alternative one: UNIX domain sockets. Instead of binding to a TCP port on 127.0.0.1, the server can listen on a filesystem socket such as /run/psycosupport/server.sock (AF_UNIX). Access is then governed by ordinary file permissions, so the directory can be set to mode 0700 owned by the service user, and other local users cannot connect at all. There is no network stack involved, so nothing appears on the loopback interface for Wireshark to capture, and the server can use SO_PEERCRED to verify the connecting process’s UID rather than trusting a username string sent in the message.

Alternative two: TLS 1.3 with a self-signed certificate. If TCP is retained, the connection should be wrapped in TLS 1.3 using Python's ssl module, with a locally generated certificate that the client pins or trusts explicitly. TLS 1.3 negotiates ephemeral keys per session (forward secrecy) and encrypts with an AEAD cipher such as AES-256-GCM or ChaCha20-Poly1305, so every record has a unique nonce, is authenticated against tampering, and shows no letter-frequency structure. Mutual TLS, where the server also requires a client certificate, would additionally replace the self-reported username with a cryptographic identity.

Either option, replacing the hand-built affine scheme with vetted primitives, removes the attack surface demonstrated here. The two can also be combined by running TLS over a UNIX socket if defense in depth is required.